Tutorial Questions - Probability Distributions

1. Bernoulli Distribution

A web application records whether a user login attempt is successful or unsuccessful. The probability of a successful login is 0.95.

Answer the following:
(a) Find the probability of a successful login.
(b) Find the mean.
(c) Find the variance.
(d) Interpret the mean and variance.

2. Binomial Distribution

A student attempts to log in to a university portal 10 times. The probability of a successful login on each attempt is 0.9.

Answer the following:
(a) Find the probability of exactly 8 successful logins.
(b) Find the mean.
(c) Find the variance.
(d) Interpret the mean and variance.

3. Geometric Distribution

A network engineer repeatedly sends packets until the first successful transmission. The probability of success in each transmission is 0.8.

Answer the following:
(a) Find the probability that the first success occurs on the 3rd transmission.
(b) Find the mean.
(c) Find the variance.
(d) Interpret the mean and variance.

4. Negative Binomial Distribution

A server repeatedly attempts to establish secure connections. The probability of success in each attempt is 0.8.

Answer the following:
(a) Find the probability that the 4th successful connection occurs on the 6th attempt.
(b) Find the mean.

(c) Find the variance.
(d) Interpret the mean and variance.

5. Poisson Distribution

A software company observes an average of 3 software bugs reported per day.

Answer the following:
(a) Find the probability of exactly 2 bugs tomorrow.
(b) Find the mean.
(c) Find the variance.
(d) Interpret the mean and variance.

6. Uniform Distribution

A web server responds to client requests in a time equally likely between 2 and 6 seconds.

Answer the following:
(a) Find the probability that the response time is between 3 and 5 seconds.
(b) Find the mean.
(c) Find the variance.
(d) Interpret the mean and variance.

7. Exponential Distribution

A cloud storage service experiences one download failure every 30 minutes on average.

Answer the following:
(a) Find the probability that the next download failure occurs within 15 minutes.
(b) Find the mean.
(c) Find the variance.
(d) Interpret the mean and variance.

8. Normal Distribution

A web application has response times that are normally distributed with a mean of 200 ms and a standard deviation of 20 ms.

Answer the following:
(a) Find the probability that the response time lies between 180 ms and 220 ms.
(b) Find the mean.
(c) Find the variance.
(d) Interpret the mean and variance.